

Positioning for the Nuss procedure: avoiding brachial plexus injury

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Summary

Background: In 1987, Nuss developed a minimally invasive technique for the treatment of pectus excavatum. The procedure has had excellent results with minimal complications. Anesthetic concerns include risks of cardiothoracic injuries, dysrhythmias, pneumothorax, pleural effusions, and hemorrhage. In addition, we identified a risk of brachial plexus injury that can occur secondary to patient positioning. We report our experience with the Nuss procedure and modified patient positioning.

Methods: A retrospective chart review was performed on 95 children or adolescents who underwent a Nuss procedure at the Children's Hospital in Denver, CO, USA. The first group of patients (58) was positioned with their arms extended above their heads at a 90° angle at the shoulder. The second group of patients (37) was positioned in an arthroscopy sling in a neutral position utilizing a right angled suspension device attached to the operating room bed.

Results: In the first group of patients, three of 58 (5.2%) children had transient brachial plexus injuries. To date, in the second group of patients, there have been no brachial plexus injuries using the suspension device.

Conclusions: We found that patients are at risk of transient brachial plexus injury during the Nuss procedure due to positioning of the arms. A readily available device that permits the extremity to be maintained in a neutral position but also allows intraoperative manipulation during critical aspects of the Nuss procedure eliminates transient brachial plexus palsy.

Keywords: pectus excavatum; peripheral nerve injuries; patient positioning; Nuss technique

Introduction

Pectus excavatum, a relatively common congenital chest wall deformity in children, is a depression of

the sternum that commonly starts at the angle of Louis, and is deepest at the xiphisternal junction (1–5). The classic surgical repair of pectus excavatum is the Ravitch repair, which entails subperichondrial resection of the deformed cartilages and a sternal osteotomy (6). However, in 1987, Dr Donald Nuss developed a minimally invasive technique for

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treatment of pectus excavatum (1). Subsequently, the Nuss procedure was modified so that a curved steel bar is placed through an endoscopically guided retrosternal tunnel between incisions in the mid-axillary line bilaterally at the level of the T4–5 interspace (7).

The anesthetic technique for the Nuss procedure has been limited to general anesthesia as the chest is entered endoscopically and positive pressure ventilation is desired. To supplement a general anesthetic, most pectus patients also receive a thoracic epidural placed prior to induction. The epidural can then be utilized intraoperatively to decrease anesthetic requirements and postoperatively to provide superior postoperative pain control compared with PCA narcotics (8). Although the Nuss procedure has enjoyed great success with minimal complications (2), several anesthetic concerns remain. These include risks of cardiac perforation, dysrhythmias, pneumothorax, pleural/pericardial effusions and hemorrhage (3,4,9,10). In addition, there appears to be a risk of brachial plexus injury secondary to patient positioning.

During the Nuss procedure, the patient position required, entails abduction of the arms in order to allow the surgeon access to the lateral chest wall. At critical times during the surgery, this upper extremity abduction may exceed 90° to allow for endoscopic camera insertion and bar placement. The purpose of this paper is to compare two different patient positions utilized during the Nuss procedure to determine the incidence of brachial nerve injury in each group. In addition, patient's age and surgical times were evaluated to determine their possible impact on the incidence of nerve palsy.

Methods

The study was approved by the Institutional Review Board for the department of research compliance, University of Colorado (protocol no. 04-0635). Records of 95 children and adolescents who underwent the Nuss procedure at The Childrens Hospital, Denver, CO, USA from 1999 to 2004 were reviewed retrospectively. Parameters recorded included patient age, patient position on operating room table (group 1 and group 2 as described below), length of procedure and the presence of postoperative brachial plexus palsy. Patients were classified as having postoperative brachial plexus palsy if they experi-

enced new-onset symptoms of either pain and/or paresthesias of the upper extremity following the Nuss procedure. The charts of this subgroup of patients were also reviewed in order to determine the duration of brachial plexus palsy. Statistical comparisons between groups one and two were made using Fischer's exact test or Student's *t*-test where appropriate.

The first group of patients (58) was positioned with their arms above their heads at a 90° angle of abduction at the shoulder and flexed at 45–60° at the elbow. (Figure 1) The second group of patients (37) was positioned in an arthroscopy sling suspended from a right angle suspension device (Figure 2). The arm was first wrapped in cast padding, free of all IVs and monitors. Next the sling was applied and secured with adhesive dressing. The extremity was then suspended on the right angle device which was secured to the operating room table at a slightly less than 90° angle of abduction at the shoulder. The upper arm and elbow remain in a neutral position to the patient's body.

Results

In the first group of patients, three of the 58 children (5.2%) had transient brachial plexus injuries. To



Figure 1
Original positioning with left arm extended 90° at shoulder joint and elbow flexed 45–90°.



Figure 2
Modified position with left arm in arthroscopy sling suspended from right angle suspension device.

date, in the second group of patients who were positioned in the arthroscopy sling device ($n = 37$) there have been no brachial plexus injuries. (Table 1) Data are expressed as the mean $\pm$ SE of the mean where appropriate. Statistical comparisons of nominal data were made using Student's *t*-test. Proportional data were compared using Fischer's exact test. Because of the small number of patients with injuries, statistically significant differences in age or operative times could not be demonstrated between the injury group and either group one or group two patients.

All brachial plexus injuries were transient, lasting 2 days to 5 weeks (mean 2 days; range 1–5 weeks). Symptoms included pain and/or paresthesias of the left arm. All patients who sustained palsys were given a complete neurological examination to delineate the extent of nerve involvement. In addition,

Table 1
Patient data

	Group no. 1	Group no. 2	P-value
Number of patients	$n = 58$	$n = 37$	
Age (years)			
Mean ($\pm$ SEM)	13.1 ± 0.6	12.1 ± 0.6	0.237 ^a
Median	14	12	na
Range	4–20	6–25	na
Operative time (min) ($\pm$ SEM)	101.7 ± 5.2	103.3 ± 5.4	0.829 ^a
Number of injuries	3	0	
% Complications	5.2%	0%	<0.0001 ^b

^a*t*-test; ^bFishers exact test.

patients were followed with serial exams until complete resolution of injury was documented. Conservative therapy consisted of nonsteroidal anti-inflammatory medications for pain and physical therapy.

The symptoms of brachial nerve palsy experienced by the three patients in group one were limited to upper arm numbness and decreased motor function. The decreased sensation occurred in the patient's upper arm extending to the elbow. The motor weakness associated with the paresthesia involved shoulder abduction, extension and external rotation, and elbow flexion. This type of injury is most closely associated with the distribution of nerve roots C5–7 (11).

Discussion

Pectus excavatum is the most common childhood chest deformity that is treated surgically. A minimally invasive surgical technique, the Nuss procedure, has produced excellent results with minimal complications (12). We identified an additional complication of the Nuss procedure, transient brachial plexus injury that appears to result from patient positioning. Because the endoscopic camera and the steel bar are both placed in the lateral chest wall, the surgeon requires access to this area. Therefore, the patient must be positioned with upper extremities abducted at the shoulder. At times during the surgery, the surgeon may need this angle of upper extremity abduction to exceed 90° and we believe this results in stretching and compression of the brachial plexus (13).

Brachial plexus injuries are the second most common type of postoperative nerve injuries (13).

The risk factors for brachial plexus lesions during anesthesia include improper positioning of patients, extreme variations of body mass index and certain anatomical anomalies (14). The plexus is uniquely susceptible because of the long course it takes between its two points of fixation: the vertebral fascia proximally and the axillary fascia distally. In addition, there are several bony structures in close proximity to the plexus, which can serve to compress the nerve. These bony structures include the first rib, the clavicle and the head of the humerus. Finally, peripheral nerve injuries in general, and brachial plexus injuries in particular are more likely to occur during surgery because in anesthetized patients there is a decrease in normal resting muscle tone which allows patients to assume nonphysiologic positions. These unnatural positions can lead to nerve ischemia when the vessels supplying the nerves become compressed (15,16). And while awake patients will move when pain or paresthesias occur to avoid nerve ischemia, the anesthetized patient is unable to do so.

Because it is known that brachial plexus injuries are a relatively common postoperative complication, and one that appears to be associated with the Nuss procedure, we hypothesize that a change in the patient positioning technique utilized during the Nuss procedure could influence the incidence of brachial plexus injury. Our findings suggest that patient positioning technique does play a role in the incidence of postoperative nerve palsy. When patients were positioned as described in group 1 and shown in Figure 1, there was a 5.2% incidence of transient, postoperative brachial nerve palsies. However, when patient position was modified as described in group 2 and shown in Figure 2, the incidence dropped to zero.

We speculate that the reason for the decreased incidence in group 2 is related to the fact that the patient's left arm, when positioned in the suspension device, allows the surgeon to move the arm up at a greater than 90° abduction at the shoulder, for brief periods, when camera or bar placement dictate such movement during the Nuss procedure. The arm then is free to swing back to its original position at less than 90° abduction for the remainder of the surgery. This is in comparison with the position utilized in group 1 patients where the arm is most likely moved up inadvertently for camera and bar placement, but

is not free to swing back into its original position. We believe this leads to excess stretching and compression of the brachial plexus and resultant nerve injury.

Of interest, the findings may also suggest that patients with increased age and greater surgical times may also be more susceptible to peripheral nerve injuries. Of the three patients in this review who did experience postoperative peripheral nerve injuries, all were older (mean age of 18.3 years), than their cohorts in either group 1 or group 2. Similarly, operating times were longer in the group of patients who experienced nerve injuries (188.3 min) compared with patients in group 1 and 2 who did not experience any injury. Because of the small number of patients, however, these differences are not significant and additional studies would need to be completed in order to determine if these relationships exists.

In conclusion, patients undergoing the Nuss procedure for pectus excavatum repair may be positioned in a readily available suspension device, which allows the upper extremity to be moved out of the surgical field during critical moments in the procedure and then swing back into a neutral position for the remainder of the surgery. This positioning technique eliminates the occurrence of brachial plexus injury.

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